

Inverse Filtering

Inverse Filtering is an image restoration technique used to recover the original image from a blurred or degraded image by reversing the effect of the degradation function.

It attempts to remove blur mathematically.

Simple Definition

Inverse filtering restores an image by dividing the degraded image by the degradation function.

Main Idea

If the degradation process is known, its effect can be reversed to estimate the original image.

Degradation Model

The degraded image is:

$$G(u, v) = H(u, v)F(u, v) + N(u, v)$$

Where:

- $G(u, v)$ = degraded image
 - $H(u, v)$ = degradation function
 - $F(u, v)$ = original image
 - $N(u, v)$ = noise
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Inverse Filtering Formula

Ignoring noise:

$$F(u, v) = \frac{G(u, v)}{H(u, v)}$$

This restores the image by removing the degradation effect.

Working Process

1. Convert image to frequency domain using Fourier Transform
 2. Apply inverse filter
 3. Recover estimated original image
 4. Convert back to spatial domain
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Problem with Inverse Filtering

If:

$$H(u, v) = 0$$

or very small,

division causes:

- instability
- huge noise amplification

So inverse filtering is very sensitive to noise.

Advantages

- Simple mathematical method
 - Effective when degradation is known exactly
 - Good for blur removal in noiseless conditions
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Disadvantages

- Highly sensitive to noise
 - Poor performance when degradation function has zeros
 - Not suitable for real noisy images
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Applications

Used in:

- Deblurring images
- Astronomical imaging

- Medical imaging
- Restoration research

Inverse Filtering vs Wiener Filtering

Feature	Inverse Filtering	Wiener Filtering
Noise Handling	Poor	Good
Complexity	Simple	More complex
Stability	Less stable	More stable

Homomorphic Filtering

Homomorphic Filtering is an image enhancement technique used to simultaneously:

- improve image contrast
- normalize brightness
- reduce uneven illumination

It works in the frequency domain.

Simple Definition

Homomorphic filtering separates illumination and reflectance components of an image and enhances the image by reducing illumination effects.

Main Idea

An image consists of two components:

$$f(x, y) = i(x, y) \times r(x, y)$$

Where:

- $i(x, y)$ = illumination component
 - $r(x, y)$ = reflectance component
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Illumination Component

- Represents lighting
 - Usually low-frequency
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Reflectance Component

- Represents object details
 - Usually high-frequency
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Goal of Homomorphic Filtering

- Suppress low frequencies (uneven lighting)

- Enhance high frequencies (details and edges)
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Working Process

Step 1: Log Transformation

Convert multiplication into addition:

$$\ln f(x, y) = \ln i(x, y) + \ln r(x, y)$$

Step 2: Fourier Transform

Convert image into frequency domain.

Step 3: Apply High-Pass Filter

- Reduce illumination component
 - Enhance reflectance component
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Step 4: Inverse Fourier Transform

Convert back to spatial domain.

Step 5: Exponential Operation

Restore final enhanced image.

Advantages

- Enhances contrast
 - Removes uneven illumination
 - Sharpens image details
 - Improves visual quality
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Disadvantages

- Computationally complex

- May amplify noise
- Parameter selection is important

Applications

Used in:

- Medical imaging
- Satellite imaging
- Fingerprint enhancement
- Face recognition
- Low-light image enhancement

Homomorphic Filtering vs Normal Filtering

Feature	Homomorphic Filtering	Normal Filtering
Purpose	Illumination correction + enhancement	Noise removal/smoothing
Domain	Frequency domain	Spatial/Frequency
Handles Lighting	Yes	Usually No

Degradation Function

The Degradation Function describes how an original image gets distorted or degraded during image acquisition, transmission, or processing.

It models the effect of:

- blur
- noise
- motion
- atmospheric disturbance
- sensor imperfections

on an image.

Simple Definition

A degradation function mathematically represents the process that causes image quality loss.

Main Idea

The observed image is usually not perfect because some unwanted effects modify the original image.

The degradation function explains this transformation.

Mathematical Model

The degraded image model is:

$$g(x, y) = h(x, y) * f(x, y) + \eta(x, y)$$

Where:

- $f(x, y)$ = original image
- $h(x, y)$ = degradation function
- $*$ = convolution
- $\eta(x, y)$ = noise
- $g(x, y)$ = degraded image

Frequency Domain Representation

Using Fourier Transform:

$$G(u, v) = H(u, v)F(u, v) + N(u, v)$$

Where:

- $F(u, v)$ = original image frequency
 - $H(u, v)$ = degradation function
 - $N(u, v)$ = noise
 - $G(u, v)$ = degraded image
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Causes of Image Degradation

1. Motion Blur

Occurs when camera or object moves during capture.

2. Defocus Blur

Occurs when camera focus is incorrect.

3. Atmospheric Turbulence

Common in satellite and astronomical imaging.

4. Noise

Added due to sensors or transmission errors.

Examples:

- Gaussian noise
 - Salt-and-pepper noise
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Purpose of Degradation Function

It helps in:

- image restoration
 - noise removal
 - deblurring
 - improving image quality
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Image Restoration

If degradation function is known, restoration techniques like:

- inverse filtering
- Wiener filtering

can recover the original image.

Advantages

- Helps model image distortion mathematically
 - Useful in restoration algorithms
 - Improves image analysis accuracy
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Disadvantages

- Exact degradation function may be difficult to determine
 - Complex real-world degradation
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Applications

Used in:

- Medical imaging
- Satellite imaging
- Astronomy
- CCTV enhancement
- Computer Vision